Lab 1: NVMeVirt with ZNS vs. Traditional FTL

1. two namespaces running (conventional vs. ZNS) and their Identify info
   - conventional

```
Node             SN                   Model                                    Namespace Usage                      Format           FW Rev
---------------- -------------------- ---------------------------------------- --------- -------------------------- ---------------- --------
/dev/nvme0n1     CSL_Virt_SN_01       CSL_Virt_MN_01                           1          1.00  GB /   1.00  GB    512   B +  0 B   CSL_002
```

```
NVME Identify Namespace 1:
nsze    : 0x1de099
ncap    : 0x1de099
nuse    : 0x1de099
nsfeat  : 0
nlbaf   : 6
flbas   : 0
mc      : 0
dpc     : 0
dps     : 0
nmic    : 0
rescap  : 0
fpi     : 0
dlfeat  : 0
nawun   : 0
nawupf  : 0
nacwu   : 0
nabsn   : 0
nabo    : 0
nabspf  : 0
noiob   : 0
nvmcap  : 0
mssrl   : 0
mcl     : 0
msrc    : 0
anagrpid: 0
nsattr  : 0
nvmsetid: 0
endgid  : 0
nguid   : 00000000000000000000000000000000
eui64   : 0000000000000000
lbaf  0 : ms:0   lbads:9  rp:0x2 (in use)
lbaf  1 : ms:8   lbads:9  rp:0x2
lbaf  2 : ms:16  lbads:9  rp:0x2
lbaf  3 : ms:0   lbads:12 rp:0
lbaf  4 : ms:8   lbads:12 rp:0
lbaf  5 : ms:64  lbads:12 rp:0
lbaf  6 : ms:128 lbads:12 rp:0
```

- ZNS

```
Node             SN                   Model                                    Namespace Usage                      Format           FW Rev
---------------- -------------------- ---------------------------------------- --------- -------------------------- ---------------- --------
/dev/nvme0n1     CSL_Virt_SN_01       CSL_Virt_MN_01                           1          2.15  GB /   2.15  GB    512   B +  0 B   CSL_002
```

```
ZNS Command Set Identify Namespace:
zoc       : 0
ozcs      : 0
mar       : 0x7
mor       : 0x7
rrl       : 0
frl       : 0
rrl1      : 0
rrl2      : 0
rrl3      : 0
frl1      : 0
frl2      : 0
frl3      : 0
lbafe  0: zsze:0x80000 zdes:0 (in use)
lbafe  1: zsze:0x0 zdes:0
lbafe  2: zsze:0x0 zdes:0
lbafe  3: zsze:0x0 zdes:0
lbafe  4: zsze:0x0 zdes:0
lbafe  5: zsze:0x0 zdes:0
lbafe  6: zsze:0x0 zdes:0
```

2. benchmark runs on both namespaces.
   - FTL + F2FS

```
fio-3.28
Starting 4 processes
Jobs: 2 (f=2): [_(2),f(2)][-.-%][w=99.6MiB/s][w=25.5k IOPS][eta 00m:00s]
f2fs_test: (groupid=0, jobs=1): err= 0: pid=4178: Tue Oct 21 11:51:52 2025
  write: IOPS=42.5k, BW=166MiB/s (174MB/s)(100MiB/603msec); 0 zone resets
    clat (nsec): min=1388, max=248700, avg=22023.50, stdev=7133.20
     lat (nsec): min=1416, max=248733, avg=22173.68, stdev=7150.26
    clat percentiles (nsec):
     |  1.00th=[ 1544],  5.00th=[19584], 10.00th=[20608], 20.00th=[20864],
     | 30.00th=[20864], 40.00th=[21120], 50.00th=[21120], 60.00th=[21376],
     | 70.00th=[21632], 80.00th=[22400], 90.00th=[25728], 95.00th=[37120],
     | 99.00th=[42752], 99.50th=[52480], 99.90th=[74240], 99.95th=[76288],
     | 99.99th=[81408]
  lat (usec)   : 2=3.81%, 4=0.73%, 10=0.07%, 20=0.81%, 50=94.00%
  lat (usec)   : 100=0.57%, 250=0.01%
  cpu          : usr=0.00%, sys=0.00%, ctx=1, majf=0, minf=13
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,25600,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
f2fs_test: (groupid=0, jobs=1): err= 0: pid=4179: Tue Oct 21 11:51:52 2025
  write: IOPS=7582, BW=29.6MiB/s (31.1MB/s)(100MiB/3376msec); 0 zone resets
    clat (nsec): min=674, max=86392, avg=1557.50, stdev=1886.55
     lat (nsec): min=702, max=86422, avg=1589.95, stdev=1889.29
    clat percentiles (nsec):
     |  1.00th=[  820],  5.00th=[  884], 10.00th=[  916], 20.00th=[  972],
     | 30.00th=[ 1020], 40.00th=[ 1112], 50.00th=[ 1496], 60.00th=[ 1624],
     | 70.00th=[ 1704], 80.00th=[ 1784], 90.00th=[ 1960], 95.00th=[ 2544],
     | 99.00th=[ 3408], 99.50th=[13888], 99.90th=[25984], 99.95th=[40704],
     | 99.99th=[56576]
  lat (nsec)   : 750=0.17%, 1000=25.79%
  lat (usec)   : 2=65.07%, 4=8.32%, 10=0.15%, 20=0.25%, 50=0.24%
  lat (usec)   : 100=0.02%
  cpu          : usr=0.00%, sys=1.54%, ctx=86, majf=0, minf=13
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,25600,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
```

```
f2fs_test: (groupid=0, jobs=1): err= 0: pid=4180: Tue Oct 21 11:51:52 2025
  write: IOPS=6074, BW=23.7MiB/s (24.9MB/s)(100MiB/4214msec); 0 zone resets
    clat (nsec): min=701, max=156876k, avg=7246.97, stdev=980471.18
     lat (nsec): min=732, max=156876k, avg=7278.67, stdev=980471.21
    clat percentiles (nsec):
     |  1.00th=[   796],  5.00th=[   844], 10.00th=[   868], 20.00th=[   900],
     | 30.00th=[   932], 40.00th=[   964], 50.00th=[  1020], 60.00th=[  1080],
     | 70.00th=[  1112], 80.00th=[  1160], 90.00th=[  1240], 95.00th=[  1336],
     | 99.00th=[  1624], 99.50th=[  2096], 99.90th=[ 21888], 99.95th=[ 52992],
     | 99.99th=[108032]
  lat (nsec)   : 750=0.09%, 1000=46.48%
  lat (usec)   : 2=52.88%, 4=0.25%, 10=0.11%, 20=0.08%, 50=0.06%
  lat (usec)   : 100=0.04%, 250=0.01%
  lat (msec)   : 250=0.01%
  cpu          : usr=0.00%, sys=1.35%, ctx=88, majf=0, minf=15
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,25600,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
f2fs_test: (groupid=0, jobs=1): err= 0: pid=4181: Tue Oct 21 11:51:52 2025
  write: IOPS=7582, BW=29.6MiB/s (31.1MB/s)(100MiB/3376msec); 0 zone resets
    clat (nsec): min=671, max=145117, avg=1564.44, stdev=2215.49
     lat (nsec): min=700, max=145146, avg=1597.86, stdev=2222.34
    clat percentiles (nsec):
     |  1.00th=[   828],  5.00th=[   892], 10.00th=[   924], 20.00th=[   980],
     | 30.00th=[  1032], 40.00th=[  1112], 50.00th=[  1400], 60.00th=[  1640],
     | 70.00th=[  1720], 80.00th=[  1800], 90.00th=[  1976], 95.00th=[  2512],
     | 99.00th=[  3408], 99.50th=[  5088], 99.90th=[ 31104], 99.95th=[ 41728],
     | 99.99th=[113152]
  lat (nsec)   : 750=0.09%, 1000=23.38%
  lat (usec)   : 2=67.18%, 4=8.74%, 10=0.15%, 20=0.21%, 50=0.22%
  lat (usec)   : 100=0.01%, 250=0.01%
  cpu          : usr=0.00%, sys=1.21%, ctx=85, majf=0, minf=13
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,25600,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=94.9MiB/s (99.5MB/s), 23.7MiB/s-166MiB/s (24.9MB/s-174MB/s), io=400MiB (419MB), run=603-4214msec

Disk stats (read/write):
  nvme0n1: ios=32/1623, merge=0/49577, ticks=886/32535, in_queue=33421, util=28.57%
```

- ZNS + ZoneFS

```
fio-3.28
Starting 4 processes
zonefs_seqwrite: Laying out IO file (1 file / 1024MiB)
zonefs_seqwrite: Laying out IO file (1 file / 1024MiB)
zonefs_seqwrite: Laying out IO file (1 file / 1024MiB)
zonefs_seqwrite: Laying out IO file (1 file / 1024MiB)
Jobs: 2 (f=2): [W(1),_(2),W(1)][94.4%][w=203MiB/s][w=1620 IOPS][eta 00m:01s]
zonefs_seqwrite: (groupid=0, jobs=1): err= 0: pid=3409: Mon Oct 20 22:39:51 2025
  write: IOPS=499, BW=62.4MiB/s (65.5MB/s)(1024MiB/16404msec); 0 zone resets
    clat (usec): min=260, max=43335, avg=1983.30, stdev=5408.48
     lat (usec): min=261, max=43337, avg=1999.00, stdev=5434.46
    clat percentiles (usec):
     |  1.00th=[  355],  5.00th=[  396], 10.00th=[  412], 20.00th=[  457],
     | 30.00th=[  510], 40.00th=[  545], 50.00th=[  603], 60.00th=[  693],
     | 70.00th=[  840], 80.00th=[ 1074], 90.00th=[ 1778], 95.00th=[ 8455],
     | 99.00th=[29492], 99.50th=[32375], 99.90th=[38536], 99.95th=[40633],
     | 99.99th=[43254]
   bw (  KiB/s): min=36023, max=115200, per=24.09%, avg=61599.19, stdev=23102.62, samples=32
   iops        : min=  281, max=  900, avg=481.03, stdev=180.49, samples=32
  lat (usec)   : 500=27.43%, 750=37.10%, 1000=13.38%
  lat (msec)   : 2=13.27%, 4=2.75%, 10=1.27%, 20=0.92%, 50=3.89%
  cpu          : usr=0.35%, sys=17.15%, ctx=7898, majf=0, minf=13
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,8192,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
zonefs_seqwrite: (groupid=0, jobs=1): err= 0: pid=3410: Mon Oct 20 22:39:51 2025
  write: IOPS=527, BW=65.9MiB/s (69.1MB/s)(1024MiB/15538msec); 0 zone resets
    clat (usec): min=281, max=53731, avg=1872.28, stdev=5139.47
     lat (usec): min=283, max=53732, avg=1889.11, stdev=5172.18
    clat percentiles (usec):
     |  1.00th=[  343],  5.00th=[  392], 10.00th=[  412], 20.00th=[  453],
     | 30.00th=[  510], 40.00th=[  545], 50.00th=[  594], 60.00th=[  701],
     | 70.00th=[  840], 80.00th=[ 1057], 90.00th=[ 1745], 95.00th=[ 5800],
     | 99.00th=[28705], 99.50th=[31327], 99.90th=[37487], 99.95th=[41681],
     | 99.99th=[53740]
   bw (  KiB/s): min=39680, max=138219, per=26.44%, avg=67596.60, stdev=29565.28, samples=30
   iops        : min=  310, max= 1079, avg=527.87, stdev=230.88, samples=30
  lat (usec)   : 500=28.41%, 750=35.53%, 1000=13.82%
  lat (msec)   : 2=13.62%, 4=2.70%, 10=1.65%, 20=0.76%, 50=3.49%
  lat (msec)   : 100=0.02%
  cpu          : usr=0.44%, sys=18.52%, ctx=7902, majf=0, minf=14
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
     issued rwts: total=0,8192,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
zonefs_seqwrite: (groupid=0, jobs=1): err= 0: pid=3411: Mon Oct 20 22:39:51 2025
  write: IOPS=550, BW=68.8MiB/s (72.1MB/s)(1024MiB/14893msec); 0 zone resets
    clat (usec): min=261, max=58955, avg=1808.95, stdev=5055.76
     lat (usec): min=262, max=58957, avg=1813.77, stdev=5056.50
    clat percentiles (usec):
     |  1.00th=[  343],  5.00th=[  392], 10.00th=[  412], 20.00th=[  457],
     | 30.00th=[  510], 40.00th=[  553], 50.00th=[  619], 60.00th=[  725],
     | 70.00th=[  881], 80.00th=[ 1123], 90.00th=[ 1647], 95.00th=[ 3982],
     | 99.00th=[29230], 99.50th=[31589], 99.90th=[39060], 99.95th=[43779],
     | 99.99th=[58983]
   bw (  KiB/s): min=38834, max=148950, per=27.65%, avg=70689.83, stdev=30104.86, samples=29
   iops        : min=  303, max= 1163, avg=552.07, stdev=235.18, samples=29
  lat (usec)   : 500=27.09%, 750=34.52%, 1000=14.89%
  lat (msec)   : 2=15.83%, 4=2.69%, 10=1.06%, 20=0.62%, 50=3.28%
  lat (msec)   : 100=0.01%
  cpu          : usr=0.34%, sys=16.51%, ctx=7916, majf=0, minf=14
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,8192,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1
zonefs_seqwrite: (groupid=0, jobs=1): err= 0: pid=3412: Mon Oct 20 22:39:51 2025
  write: IOPS=509, BW=63.7MiB/s (66.8MB/s)(1024MiB/16065msec); 0 zone resets
    clat (usec): min=241, max=48429, avg=1950.25, stdev=5341.89
     lat (usec): min=242, max=48431, avg=1957.66, stdev=5348.90
    clat percentiles (usec):
     |  1.00th=[  343],  5.00th=[  388], 10.00th=[  408], 20.00th=[  453],
     | 30.00th=[  502], 40.00th=[  545], 50.00th=[  603], 60.00th=[  709],
     | 70.00th=[  848], 80.00th=[ 1074], 90.00th=[ 1778], 95.00th=[ 6652],
     | 99.00th=[30016], 99.50th=[32375], 99.90th=[37487], 99.95th=[40633],
     | 99.99th=[48497]
   bw (  KiB/s): min=39011, max=155337, per=25.51%, avg=65226.12, stdev=26824.79, samples=32
   iops        : min=  304, max= 1213, avg=509.44, stdev=209.54, samples=32
  lat (usec)   : 250=0.01%, 500=29.14%, 750=34.08%, 1000=14.15%
  lat (msec)   : 2=13.83%, 4=2.84%, 10=1.32%, 20=0.89%, 50=3.74%
  cpu          : usr=0.34%, sys=16.89%, ctx=8032, majf=0, minf=14
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwts: total=0,8192,0,0 short=0,0,0,0 dropped=0,0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=250MiB/s (262MB/s), 62.4MiB/s-68.8MiB/s (65.5MB/s-72.1MB/s), io=4096MiB (4295MB), run=14893-16404msec
```

```
Disk stats (read/write):
  sda: ios=0/32578, merge=0/93, ticks=0/55273, in_queue=55360, util=83.28%
```

3. short write-up comparing NVMeVirt code paths.

項目	FTL	ZNS
核心檔案	conv_ftl.c, ssd.c	zns.c, nvme_zns.c
寫入方式	由 FTL 轉換 LBA 到 PBA ，可任意寫	必須順序寫入各 zone
管理	隱藏式 FTL 控制	Host 負責 zone 開啟/重設
API	read, write, flush	加入 zone_append, zone_reset
資料	FTL 動態映射表	Zone 內循序配置

4. modified NVMeVirt printing/logging parallelism stats.
   - 修改的程式碼:

```
bool zns_proc_nvme_io_cmd(struct nvmev_ns *ns, struct nvmev_request *req, struct nvmev_result *ret)
{
    struct nvme_command *cmd = req->cmd;
    NVMEV_ASSERT(ns->csi == NVME_CSI_ZNS);
    /*still not support multi partitions ...*/
    NVMEV_ASSERT(ns->nr_parts == 1);

    atomic_inc(&active_ios);
    update_max_active();

    switch (cmd->common.opcode) {
    case nvme_cmd_write:
    case nvme_cmd_zone_append:
        atomic_inc(&total_writes);
        if (!zns_write(ns, req, ret))
            goto out;
        break;
    case nvme_cmd_read:
        atomic_inc(&total_reads);
        if (!zns_read(ns, req, ret))
            goto out;
        break;
    case nvme_cmd_flush:
        zns_flush(ns, req, ret);
        break;
    case nvme_cmd_zone_mgmt_send:
        zns_zmgmt_send(ns, req, ret);
        break;
    case nvme_cmd_zone_mgmt_recv:
        zns_zmgmt_recv(ns, req, ret);
        break;
    default:
        NVMEV_ERROR("%s: unimplemented command: %s(%d)\n", __func__,
                nvme_opcode_string(cmd->common.opcode), cmd->common.opcode);
        break;
    }
    // 印出目前統計狀況
    printk(KERN_INFO "[ZNS] Opcode=%d | Active IOs=%d | Max Active=%d | Total W=%d | Total R=%d\n",
       cmd->common.opcode,
       atomic_read(&active_ios),
       atomic_read(&max_active_ios),
       atomic_read(&total_writes),
       atomic_read(&total_reads));
out:
    atomic_dec(&active_ios);
    return true;
}
```

   - Parallelism Stats

```
[ 4887.148483] NVMeVirt: Virtual NVMe device created
[ 4887.168677] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=4
[ 4887.173679] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=5
[ 4887.173896] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=6
[ 4887.174400] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=7
[ 4887.174714] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=8
[ 4887.175017] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=9
[ 4887.175497] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=10
[ 4887.176386] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=11
[ 4887.176715] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=12
[ 4887.176904] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=13
[ 4887.177307] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=14
[ 4887.178675] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=15
[ 4887.187338] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=16
[ 4887.187840] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=17
[ 4887.188757] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=18
[ 4887.189455] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=19
[ 4887.189707] [ZNS] Opcode=2 | Active IOs=1 | Max Active=1 | Total W=0 | Total R=20
```

5. Compare — Parallelism utilization between ZoneFS and F2FS

Filesystem	Avg Active IO (from dmesg)	Max Active	備註
F2FS (FTL)	1	1	受限於 FTL 隱藏層
ZoneFS (ZNS)	3–4	4	多 zone 同時活躍